
A formalised model of the scientific publication process

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Abstract

The scientific publishing process has during the past few years undergone considerable changes. The socio-economic structures have, however, not changed much, and many academics and librarians view the current situation as highly unsatisfactory. This has triggered a number of initiatives to set up e-print repositories and electronic peer reviewed journals, which usually offer the full text for free on the Web. Serious in-depth research studying the way the scholarly communication system is affected by the Internet is needed. In this article a formal process model of the scientific publishing process is presented (the Scientific Publication Life-Cycle Model). The model has been developed in particular to provide a basis for studying the cost implications of different business models. It describes the life-cycle of the single publication, in particular the refereed journal article, from the research leading to it and writing it, to being read by other researchers years later or used as a catalyst for practical implementation. Conclusions are drawn about the usefulness of the modelling methodology for this particular purpose as well as of future uses of the model itself. In addition to providing a basis for cost studies the model could function as a road map for different types of open access initiatives.

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Introduction

A breakdown of the costs of producing and delivering a typical refereed journal paper indicates that perhaps as much as 90 per cent of the cost consists of the actual research work preceding the writing of the paper. The research work is usually financed by public bodies and the costs are in no way recuperated through the sales of the publications (as would be the case for commercial products such as books, music CDs, movies, etc.). Even if we look only at the costs of preparing, reviewing, distributing, archiving and retrieving scientific articles, thus excluding the actual production costs of the knowledge reported on, almost all the costs are in the end borne by the universities, public libraries etc. Because of the commercial interests of one group of stakeholders, the journal publishers, which incur a very small fraction of the total life-cycle cost, the access to scientific publications is highly restricted and expensive and the process as a whole is highly inefficient (for a recent overview see Guédon, 2001). Publishing parallel electronic versions restricted to subscribers only partly remedies the situation and leaves much of the potential of the Internet untapped.

The dilemma is that it would be in the interest of the researchers and the public to have all this information accessible for free on the Internet, facilitating global access and hyperlinking of research publications. Nevertheless it is in the legitimate interest of the publishers to make a profit from selling this information, which leads to restricted circulation, password protection schemes for digital versions of traditional journals and

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bundling of electronic subscriptions to so-called “big deals” with library consortia.

Pioneering work to try out new methods of scientific publishing has already been started, usually by enthusiasts from subgroups in the scientific community. Examples are found among others in chemical science as well as mathematics (Town *et al.*, 2002; Wegner, 2001). Examples include preprint repositories, open access scholarly journals and journals offering a review process where the readers perform the peer review after a manuscript has been posted on the Web. There are, however, many psychological, legal and institutional barriers to changing the process and these have been underestimated by the pioneers and enthusiasts (Björk, 2003). Consequently only a small part of the overall volume of the scientific communication process has so far been affected by isolated efforts involving e-journals and preprint archives. A recent study made at the Swedish School of Economics and Business Administration indicates that only about 0.7 per cent of peer reviewed journals offer open access on the Web.

The SciX (open, self-organising repository for scientific information exchange) project, which is financed through the IST programme of the European Commission, aims at demonstrating that the Internet enables new business models for the scientific publishing process which are much more cost- and time-efficient to the scientific community than the current practice (SciX, 2002). The SciX project group will create services on the Web that will enable scientists as well as practitioners from the fields of architecture and construction easy and free access to relevant research publications. In addition existing publishing practices will be analysed systematically and business models for re-engineering the scientific publishing process will be developed, also taking into account the legal, social and psychological barriers to change. The model presented in this paper is one of the deliverables of the theoretical work done within the SciX project.

Aim and scope of the modelling effort

The aim of the modelling is to help us understand the scientific publishing process and

how it is affected by the Internet, in order to provide a basis for a cost and performance analysis of various alternative ways of organising it. The model can also work as a road map for positioning various new initiatives, such as e-print repositories and harvesting tools, within the overall system of scholarly communication.

The model explicitly includes the activities of all the stakeholders involved in this system, including the activities of the:

- researchers who perform the research and write the publications;
- publishers who manage and carry out the actual publication process;
- academics who participate in the process as editors and reviewers;
- libraries that archive the publications and provide access to them;
- bibliographic services which facilitate the identification and retrieval of publications;
- readers who search for, retrieve and read publications; and
- practitioners who implement the research results directly or indirectly.

In the model the unit of observation is the single publication, how it is written, edited, printed, distributed, archived, retrieved and read, and how eventually it may affect practice. The viewpoint taken is the life-cycle cost per publication. Thus at later stages all cost and time data which are collected will be translated to a per publication basis.

The aim is to clarify the process and to find a way of measuring the total life-cycle cost of a publication, since the objective of the re-engineering efforts in the SciX project is to try to investigate business models that minimise the total life-cycle costs, rather than the cost of some particular stage. This is clearly in the interest of the public authorities that finance research as well as the researchers who are primary end users. Another aim of the modelling effort is to try to clarify the functions that the demonstrator parts of SciX try to make more effective.

The model depicts publishing and value-added services using both paper and electronic formats. Pure electronic or pure paper-based publishing could be described by subsets of the model. The same goes for free publishing on the Web (“open access”), which resembles

traditional publishing, but where certain activities such as negotiating, keeping track of and invoicing subscriptions can be almost entirely left out.

The current version of the model has some limitations, which should be kept in mind. It only includes the publication and dissemination of research results in the form of publications that in the end can be printed out and studied on paper (irrespective of whether the publications are distributed on paper or electronically). Thus forms of communication such as oral communication, unstructured use of e-mail and multimedia, sharing of data sets and models, which all are essential parts of the scientific communication process, are out of scope. These could be added at a later stage, but would also add to the complexity of the model.

The model includes some activities, which would be typical for a scientific publisher publishing several journals, allowing for economies of scale. The activities of single-journal publishers could be described by a subset. The reason for including activities such as the general activities of a publisher is that these significantly influence the cost of running individual journals in the form of the general overhead costs that publishers add to the subscription prices.

How easier access to scientific publications might influence the quality of later research and industrial practice, which use these publications as input, is clearly also an important issue, but such qualitative effects of a more efficient process are very difficult to model and even more difficult to measure, and have not been attempted in this model. The same applies for the effect of the publishing on the careers of the authors, which is an important aspect for the choice of where to publish, and has created a very strong barrier for change to pure electronic journals from established "brand name" journals.

Modelling methodology

Formal process modelling techniques are frequently used in process re-engineering efforts to clarify a process and propose changes in it (for an example see Parolini, 1999). Using a formalised tool helps in communicating about

the process. The IDEF0 method was originally developed in the 1970s for the US Army to help with large procurement projects. In order to read the model the reader needs some familiarity with the IDEF0 modelling methodology. IDEF0 stands for Integration Definition for Function Modelling (NIST, 1993). The IDEF0 method is a formal graphical modelling method.

The main concepts are the activity and the flow. The flow can be used as input, output, control or mechanism. An input represents something which in an activity is consumed to produce an output. Typical inputs could be raw materials, energy, human labour, but also information when the purpose of the activity is to transform the information to provide added value. Outputs can be used as inputs to further activities, and feedback loops are possible. Activities are governed by controls. Typical examples could be laws, guidelines and instructions for carrying out an activity etc. Mechanisms, which point at activities from below, are people, organisations, machines, software, etc. which carry out the activities. The presentation of the IDEF0 diagrams is hierarchical in that diagrams on lower levels provide detailed breakdowns of those from the higher ones.

In the SciX project the Swedish School of Economics and Business Administration has used a particular software tool called BPwin for making and editing the IDEF0 model. Compared to a simple drafting tool BPwin enhances the speed and consistency of the modelling work, especially for larger models and when changes are needed.

Overall organisation of the model

The current version of the SPLC model includes 22 separate diagrams, arranged in a hierarchy up to seven levels deep. There are typically three activity boxes on each diagram, although there are a couple of diagrams with more activities and some with only two. Although official IDEF0 guidelines recommend using up to six activities per diagram, it was felt that models with three activities are easier to read and understand. There are altogether 64 activity boxes and around 200 arrows.

The overall hierarchical breakdown of the model is shown in Figure 1. Only the separate diagrams are shown. Some diagrams are further broken down into separate activities.

The diagrams in Figure 1 are numbered using the standard IDEF0 numbering scheme, which helps keep track of the hierarchical position of each diagram. The diagrams marked with an asterisk are presented and discussed.

Description of the model

The philosophy of the model is to show how science, by a combination of research with publishing, dissemination and exploitation activities, creates new scientific knowledge that improves the productivity and quality of life in society. The process ensures that research results help in the advance of our collective body of knowledge. The main stakeholders in the process are shown as mechanism arrows coming into the activity box from below, and the main drivers controlling the behaviour of the stakeholders are shown coming in from above. From an academic viewpoint the main result is new scientific knowledge. From the viewpoint of society that funds research the most important outcome is improved productivity and quality of life.

Do research, publish, study and implement the results – breakdown (A0)

The diagram in Figure 2 is crucial for understanding the life-cycle view adopted in this modelling effort. The whole life-cycle is seen as consisting of five separate stages. The “Perform the research” stage is probably the most expensive part, usually consisting of several months of work per resulting publication, but the one least affected by the re-engineering efforts facilitated by the Internet (at least directly; indirectly the effect can be substantial in terms of better quality of the research). The “Publish and disseminate the results” and “Study the results” stages constitute the main object of study in this project. The fifth activity is “Evaluate the researcher”, acting as a control mechanism of the performer of the research activity. From the perspective of the public bodies that to a large part finance research it is the efficiency of the total process, including both the production and “consumption” of publications, that should be optimised. The important thing is that in a life-cycle analysis, the cost and efficiency of both the “Publish the results activity” and the “Study the results” activity are important. Optimising only one of these may lead to an unsatisfactory solution for the total process. Here the Internet has changed the situation dramatically, as it has for information goods that can be delivered in a digital format.

Figure 1 Hierarchical breakdown of the model, only the diagrams are shown

A-0 Context Diagram

A0 Do Research, Publish, Study and Exploit the Results *

A1 Perform the Research

A2 Publish the Results *

A21 Write Manuscript

A22 Perform Publishing Activities

A221 Publish as Monograph

A222 Publish as Conference Paper

A223 Publish as Scholarly Journal Article *

A2231 Do General Publisher's Activities

A2232 Do Journal Specific Activities

A2233 Do Article and Issue Specific Activities *

A22331 Article Specific Activities *

A22332 Prepare Issue

A22333 Publish Article

A224 Publish in Miscellaneous Form

A23 Archive and Index *

A231 Make Publication Available *

A2311 Secure Access Rights and Subscription

A2312 Make Paper Publication Available

A2313 Make Electronic Copy Available

A2314 Integrate Meta Data into Search Services*

A232 Perform Value-Adding Services

A233 Archive Securely

A3 Study the Results *

A31 Find out about Publication *

A311 Search for Publication *

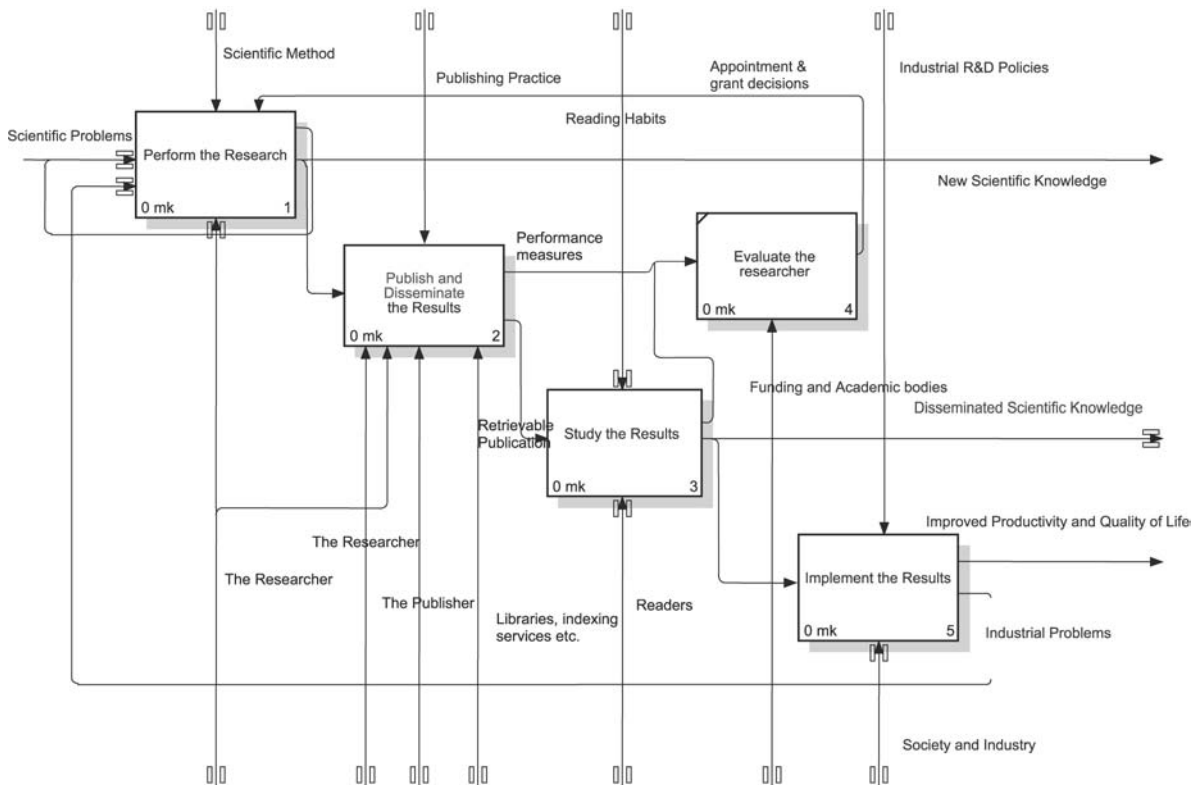
A312 Be Alerted to Publication *

A32 Retrieve Publication *

A33 Read Publication

A4 Implement the Results

Figure 2 Diagram A0 – do research, publish, study and apply the results (breakdown)



The end result of the research activity is called disseminated scientific knowledge, reflecting the viewpoint that scientific results which have been published, but which are not read by the intended readers are rather useless. In the last stage, “Implement the results”, the disseminated scientific knowledge is transformed to an improved performance of society and industry.

Publish the results (A2)

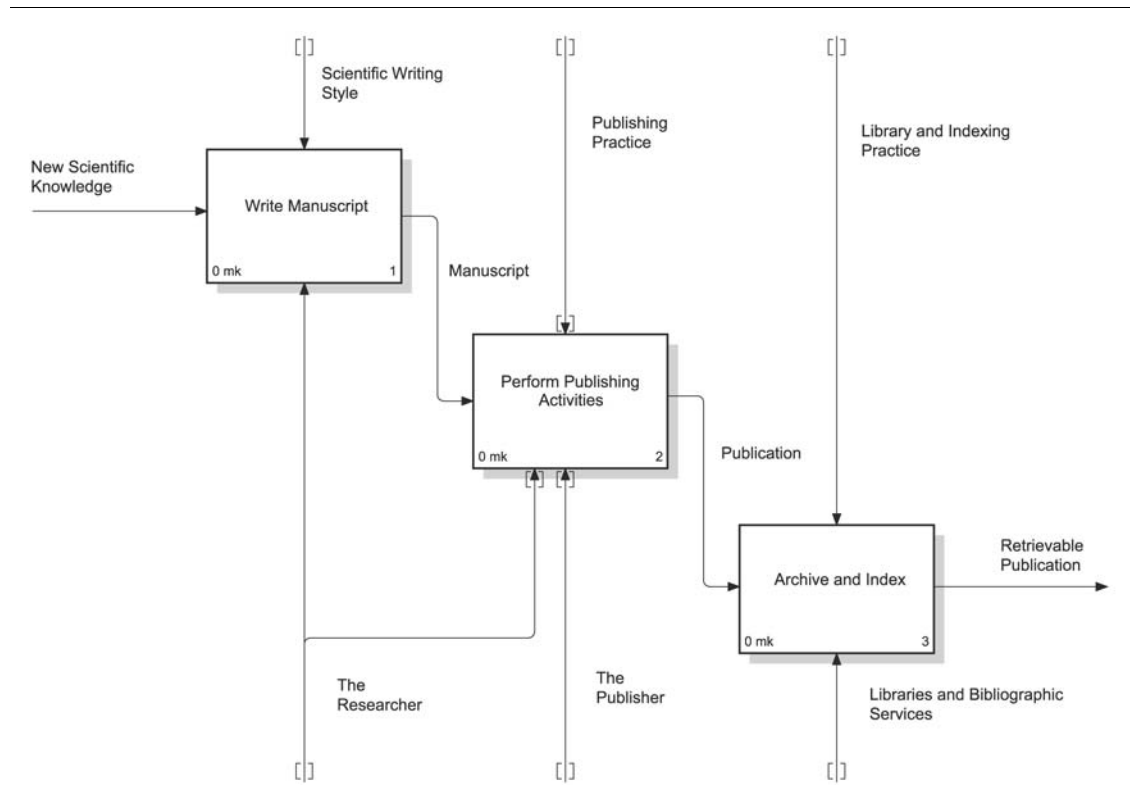
This part of the model (Figure 3) has been split up into three distinct activities, which to a large extent are carried out by different stakeholders. Based on the results of the research, the researcher writes a manuscript, which then in the next stage through a number of transformations is changed into a publication (on paper or electronic). The model will at this stage take the generic manuscript as input. Typical examples of monographs include working papers, research reports and PhD theses. Conference papers are subjected to some sort of external review either for the abstract or the full paper, and are usually presented orally in addition to the printed version. Conference proceedings are published as

one-off books or typically annually. Articles in scientific periodicals are subjected to rigorous peer review. It is important to note that periodical articles have a much higher likelihood of being referenced in bibliographical services than the other types of documents. Also journals are usually available by subscription whereas the access to monographs and conference proceedings is predominantly acquired on an individual basis. Other types of publications are, for example, popularised articles targeted for an audience of non-academics, self-publishing of multimedia content etc.

The last activity, “Archive and index”, is extremely important from a life-cycle viewpoint and involves the activities of libraries, bibliographic services, etc. to make the publication easily available to researchers and practitioners worldwide.

Of the publication types only the “Publish as journal article” has at this stage been further detailed. This is because of its relative importance in scientific publishing and also because the cost modelling effort will be concentrated there.

Figure 3 Diagram A2 – publish the results



Publish as scholarly journal article (A223)

This diagram (Figure 4) may at first sight be difficult to understand. The idea is to show all the activities which are carried out by the publishing organisation, and thus have a direct cost implication for them. This is the reason for separating activities such as “Do general publisher’s activities” and “Do journal specific activities”. Both of these demand resources, which cause overhead costs, which then are added on top of the basic variable costs caused by the processing of each individual article (under “Do article specific activities”). For instance setting up and maintaining the IT technical infrastructure for a portfolio of journals could be such an overhead item. The main pipeline in the model is, however, the input arrow “Manuscript”, which directly enters under “Do article” and “Issue specific activities”.

General publishing activities are typical for most commercial publishers and those professional associations which publish several journals. Activities can include general management and financial functions as well as the setting up of the IT technical structure for the production of journals (both on paper and

the Web). Typically such infrastructures are shared for a number of journals because of the savings in the economies of scale.

Like many of the diagrams in this model, this diagram represents a choice of viewpoint. For “Do journal specific activities” an important aspect is that commercial journals may spend a lot of money on marketing, and also on the management of subscribers (invoicing, setting up ways of checking access to electronic versions). For open access electronic journals, the latter activity is almost non-existent.

This diagram considers the two major modes for publishing scientific journals. In the paper-based world prior to the Internet, articles were as a rule bundled into issues and had to wait for publishing until the whole issue was ready. Electronic publishing does however provide the possibility of publishing each article as soon as it is ready. Today many journals are printed in both print and electronic formats but still retain the issue-based structure.

Do article specific activities (A22331)

This part of the model (Figure 5) depicts the activities carried out during the peer review

Figure 4 Diagram A223 – publish as scholarly journal article

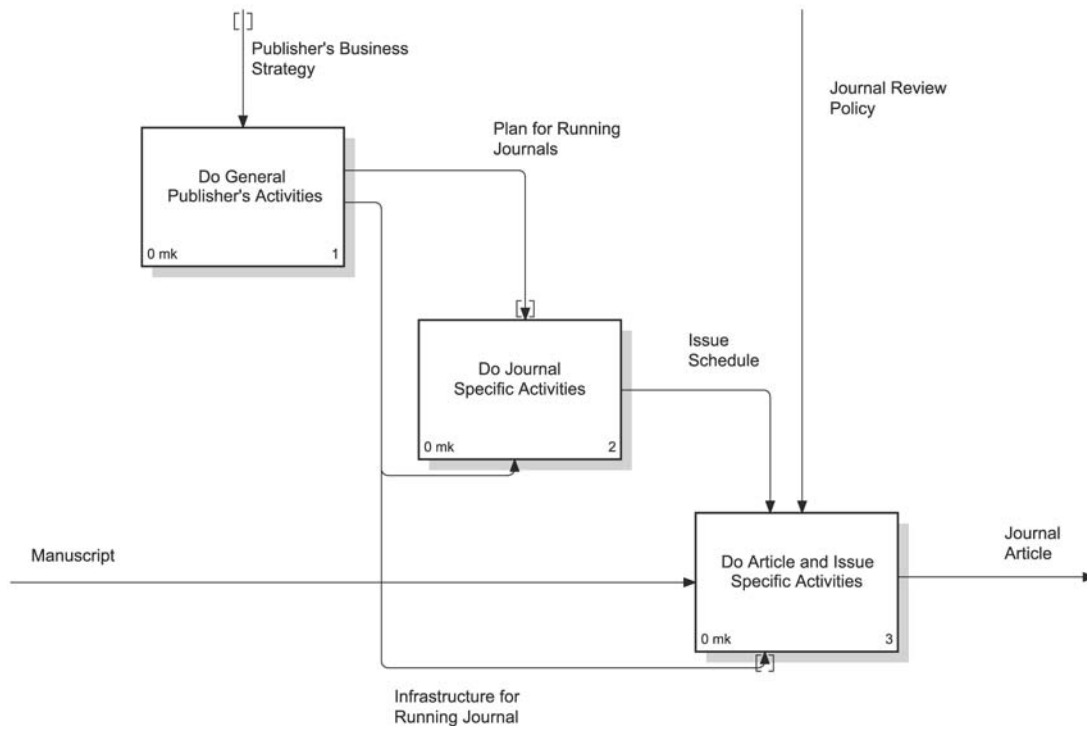
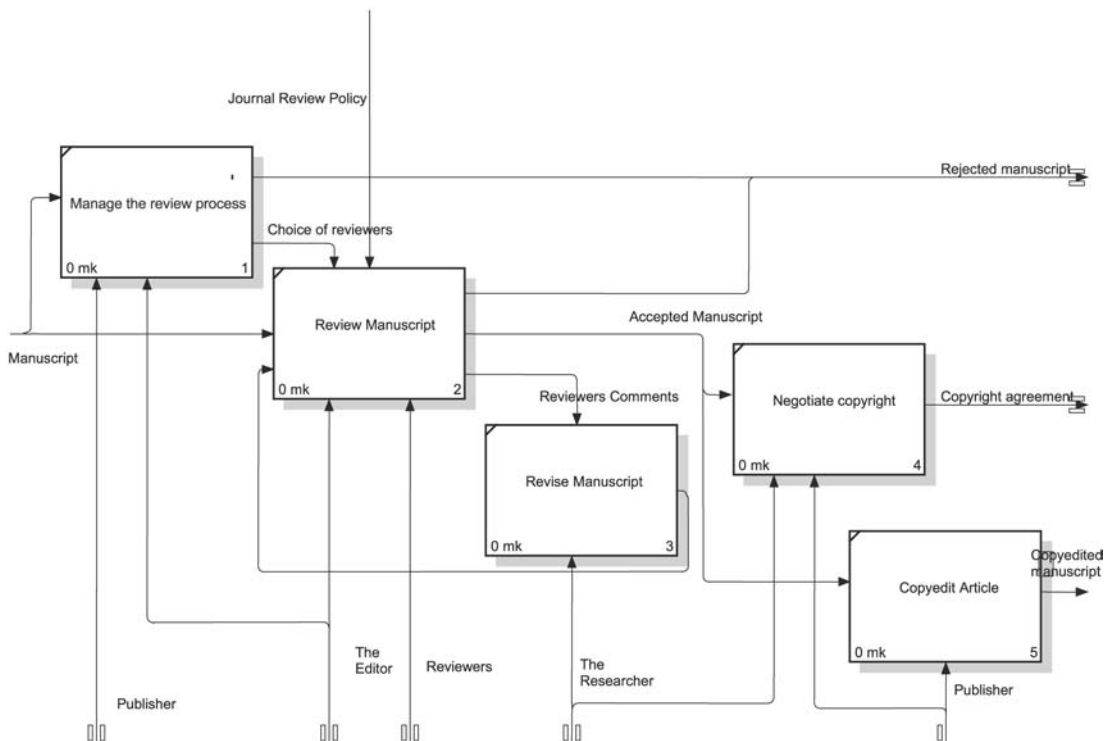


Figure 5 Diagram A2231 – do article specific activities



process as well as after the peer review to technically format the paper for printing. Note the use of a feedback mechanism, where the “Reviewers’ comments” become a control of the subsequent “Revise manuscript” activity, and where the revised manuscript is fed back into the review process. Before the advent of personal computing, copy editing used to incur considerable cost, but nowadays most researchers produce text which already is formatted according to the needs of the journal. Also the final typesetting is much easier since almost all information is acquired in a digital format. The negotiation of copyright usually takes place after the manuscript has been accepted for publication. The output is a signed copyright agreement where the author and publisher agree to the terms of publishing rights. For open access journals a copyright transfer does not usually take place since no economic rights are involved.

Once an article is accepted for publishing, it enters the “Queue for publishing”, which typically takes from half a year to a year for traditional issue and paper-based journals (the worst case the first author has experienced was three years). Waiting does not imply a direct cost, but there may be a significant opportunity cost involved from the viewpoint of the researcher and society, since the results are held back before the actual publishing. This opportunity cost is different for different domains of science. It might be low for the humanities but is usually higher in the science, technology and medical domains. In particular this is the case for IT research, where developments are extremely fast.

Archive and index (A23)

This is the part of the overall process (see Figure 6) which has traditionally to a large part been handled by research libraries with public funding. Note also that from a cost viewpoint, hundreds of libraries from all over the world have been performing the same archiving function for each paper version of an article. The primary activity here is “Make publication available”, which means that a publication is available either in print or electronically within a particular organisation (such as a university), as well as that the publication can be found in different bibliographical search services. To

“Perform value-adding services”, a third party analyses the data to calculate citation indexes, impact factors etc., or writes news bulletins about research results that practitioners can digest more easily. The “Archive” activity is currently receiving increasing attention, since the archiving of electronic versions of journals for decades implies a number of difficult problems.

Make publication available (A231)

This diagram (Figure 7) includes both the activity of making the paper publication available (placing it in the shelves of the library) and making the electronic version available. In both cases it is preceded by the longer term activity of securing subscriptions and access rights to the material, an activity which is even more visible today due to the large library consortia that negotiate deals with the large publishers. An additional value-adding activity is the integration of the metadata about the publication in databases which facilitates finding out about the existence of the publication.

There are at least two major mechanisms for making an electronic copy of a publication available. First, this can be done through standard commercial services, which necessitate that the reader or normally the local university library has secured a subscription and makes the publication visible via the university intranet. A second possibility which partly bypasses this is if the author has sent a copy of the publication to an open access e-prints repository as exemplified by the Los Alamos preprints server for physics.

Integrate metadata into search services (A2314)

Traditionally commercial indexing services have dominated the function of integrating metadata for journal articles into search services and libraries have paid subscriptions to them (Figure 8). In recent years researchers have increasingly started to use general Web search engines to identify interesting publications. An effort to overcome the quality problems related to this is the definition of the Open Archives Initiative standard for tagging scientific content material on the Web, which will enable dedicated harvesting search engines to maintain

Figure 6 Diagram A23 – archive and index

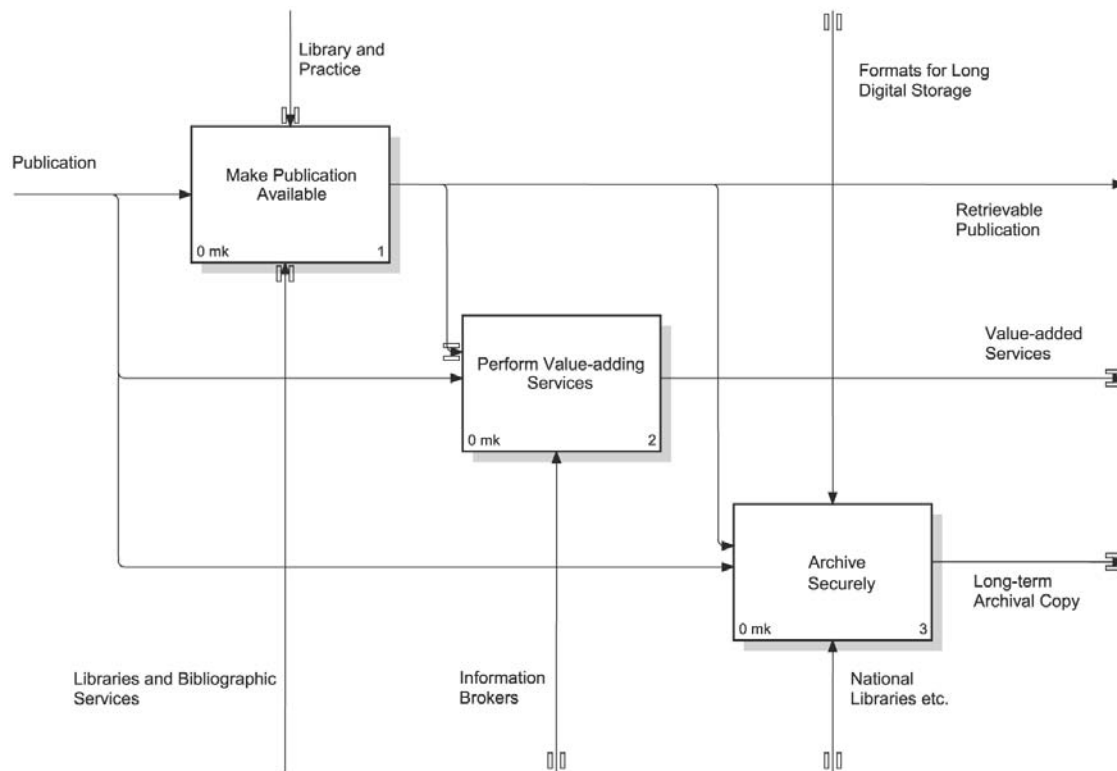


Figure 7 Diagram A231 – make publication available

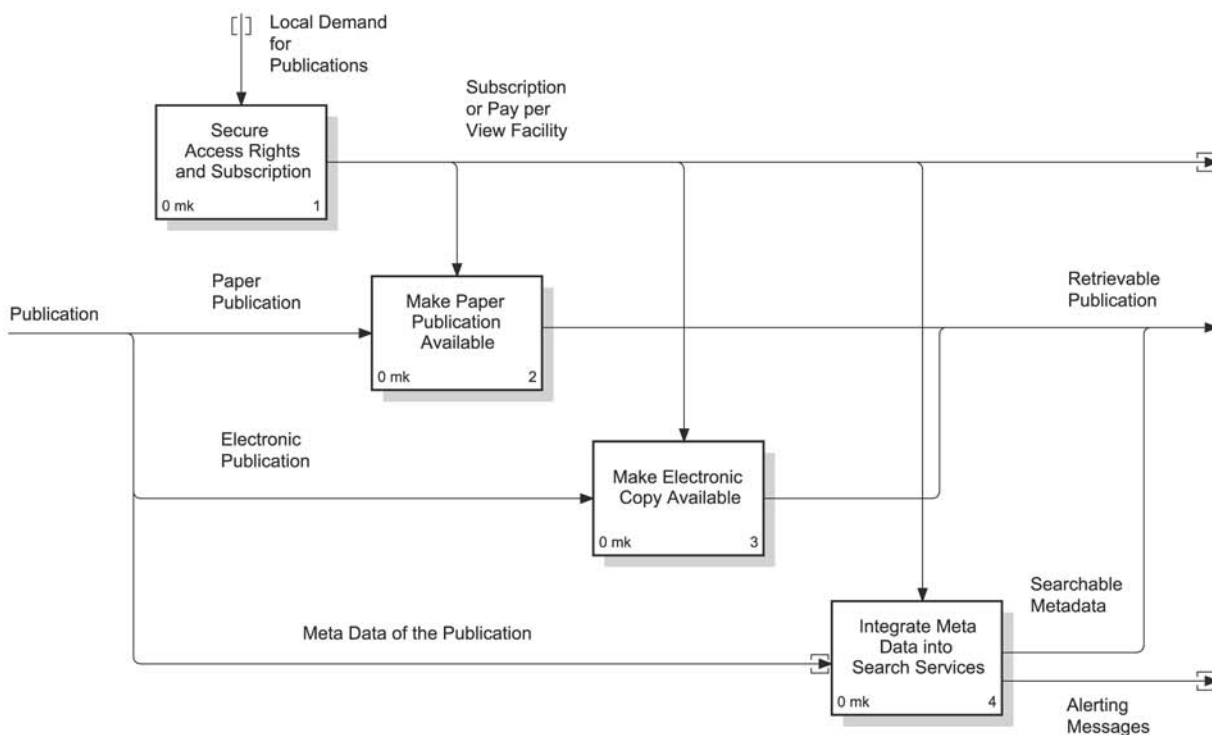
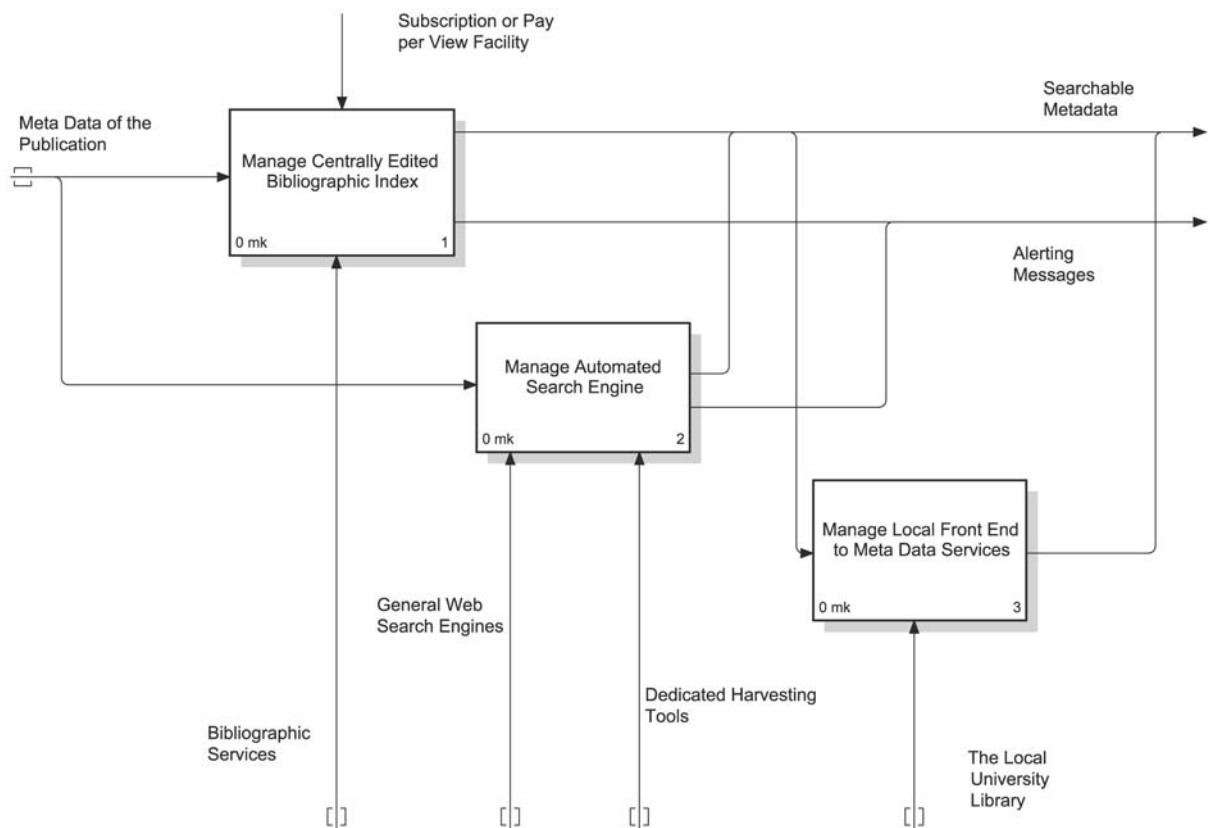


Figure 8 Diagram A2314 – integrate metadata into search services



much more focused databases of metadata about relevant publications.

A by-product of the heavy use of IT for these purposes is the possibility for readers to subscribe to services, which based on the interest profiles they define, can send them alerting messages when something they might be interested in is published.

Study the results (A3)

This diagram (Figure 9) structures the activities of the readers of scientific publications. Note that to arrive at a per publication cost the activities of individual readers all over the world and in different periods should be summed up. The “Find out about publication” activity results in the output “Metadata of interesting publication” activity (including the location from which a paper or electronic version can be retrieved). This output is used as the control of the “Retrieve publication” activity. Finally the publication is read and the scientific information in question has been disseminated. Note that researchers often self-archive

interesting publications they have read either as paper copies or today increasingly as bookmarks or in a database.

Find out about publication (A31)

This activity (Figure 10) is rather difficult to split up into alternative parallel options (for each identified article one of these is used). On this first level there is a split into two generic categories. The first one is active search where the reader is “pulling” for information. The other option is push, where the reader receives a notification through some mechanism that something interesting has been published.

Search for publication (A311)

The first modelled option of the pull variety is a traditional bibliographic search, for instance using subject words (Figure 11). Other possibilities include more unstructured Web searches using a general search engine or just browsing from one hyperlink to the other. The traditional method of physically browsing in library bookshelves is less used.

Figure 9 Diagram A3 – study the results

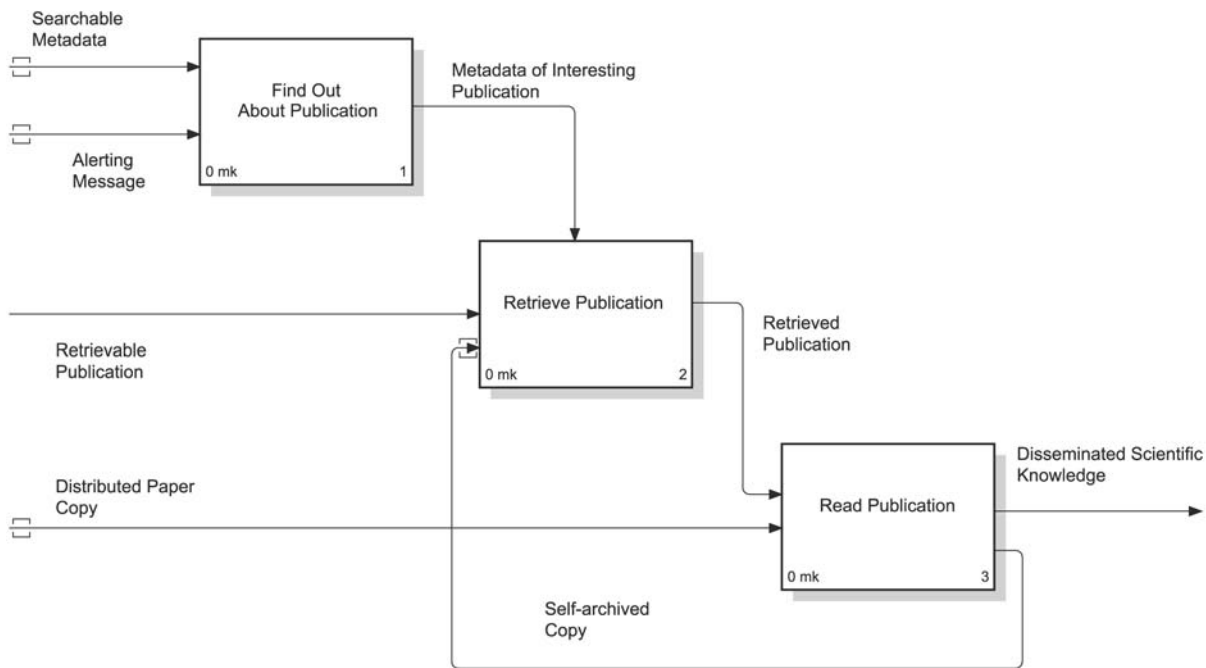
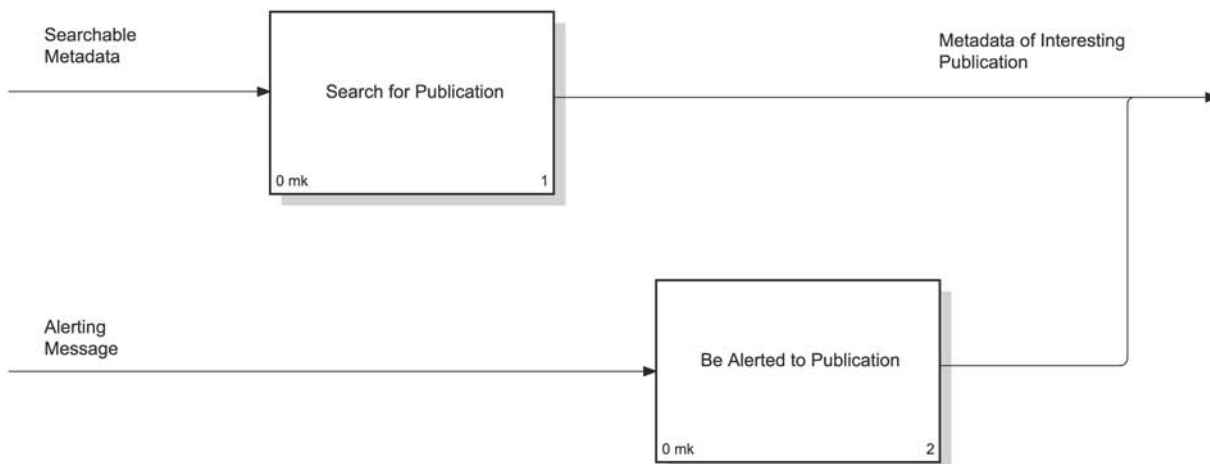


Figure 10 Diagram A31 – find out about publication



Be alerted to publication (A312)

Alerting messages can be received from a specialised service (in today's world, more likely to be software-based) or a colleague (Figure 12). The important distinction is that the bibliographic search is triggered by the researcher himself (pull) whereas the hint is coming from the outside (push). "Notice reference in other publication" is a very common way of finding out about an interesting publication. This is why it would be so convenient if all publications were available for

free on the Internet, because retrieval would only entail clicking on a hyperlink. The last option "Remember existence of publication" is less evident.

Retrieve publication (A32)

The basic split has in Figure 13 been made between the retrieval of a copy of a paper publication and one in digital form. Although the two activities modelled here may look straightforward, they might in reality become rather complex. If, for instance, the

Figure 11 Diagram A311 – search for publication

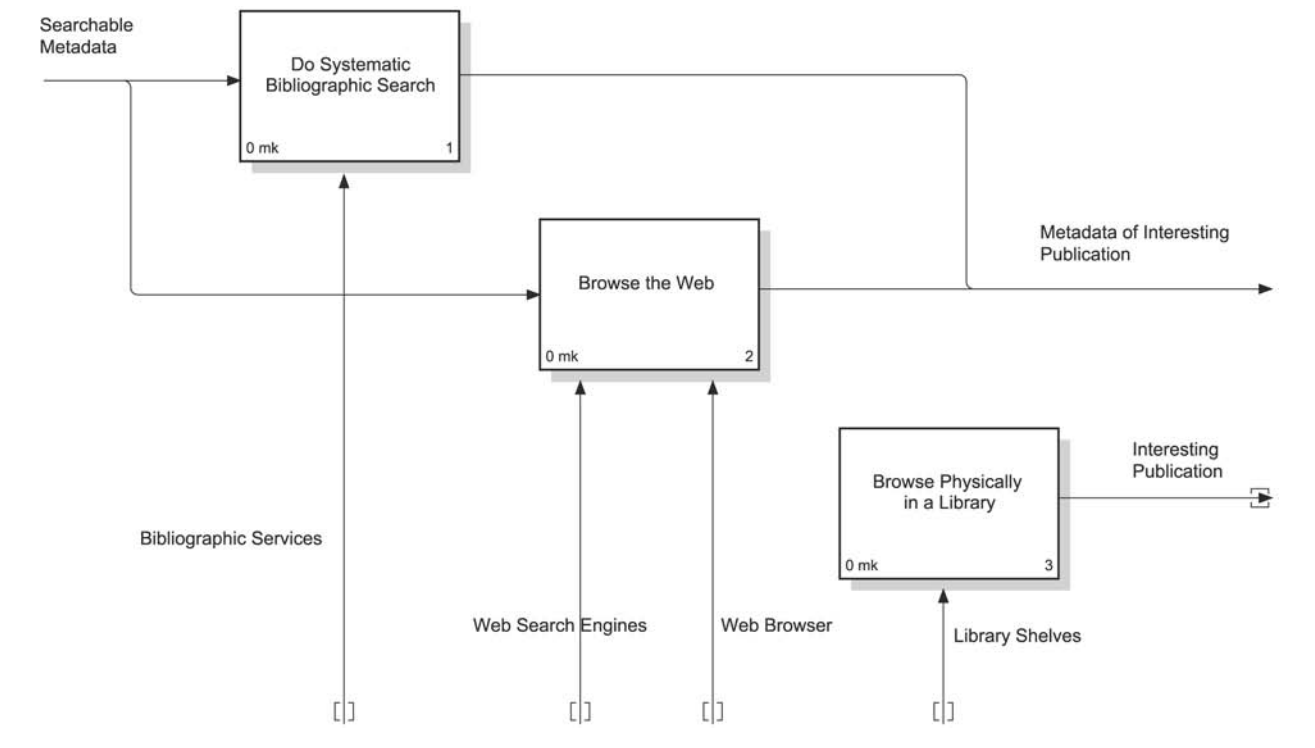


Figure 12 Diagram A312 – be alerted to publication

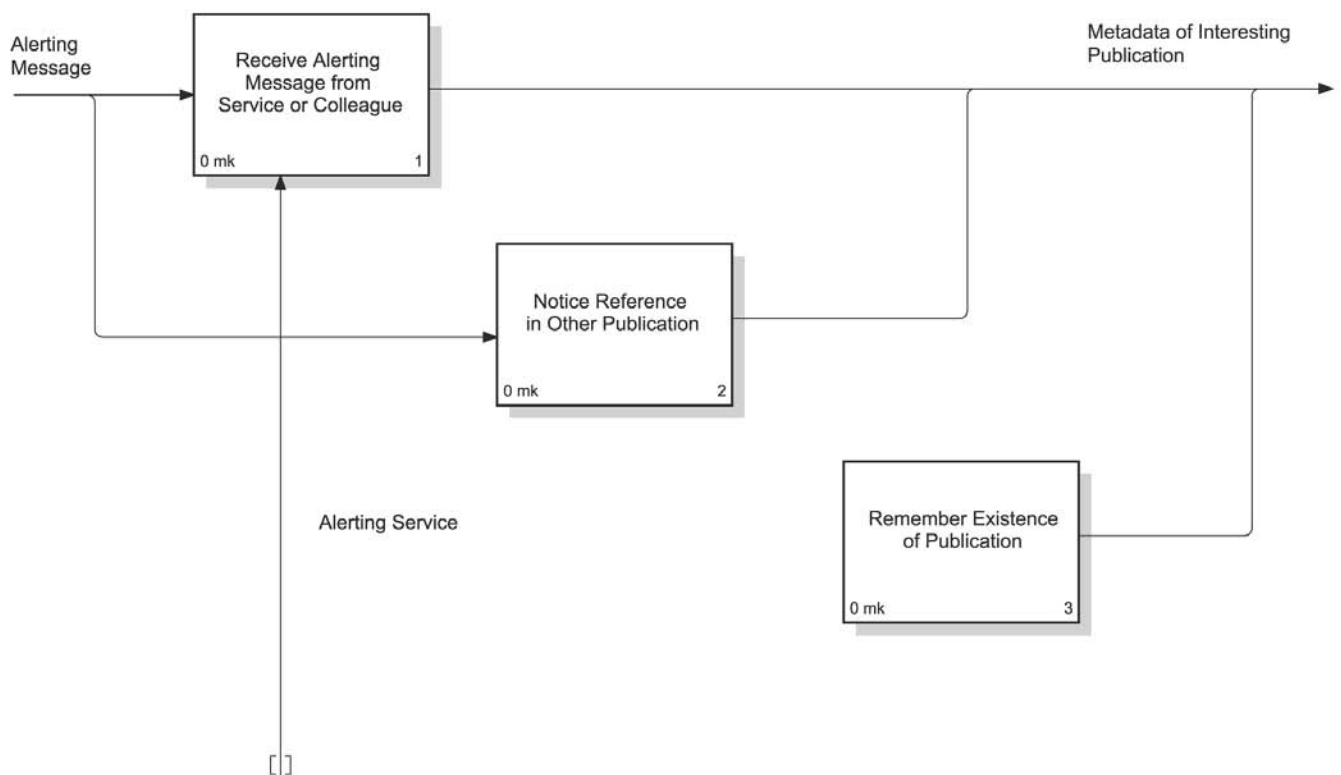
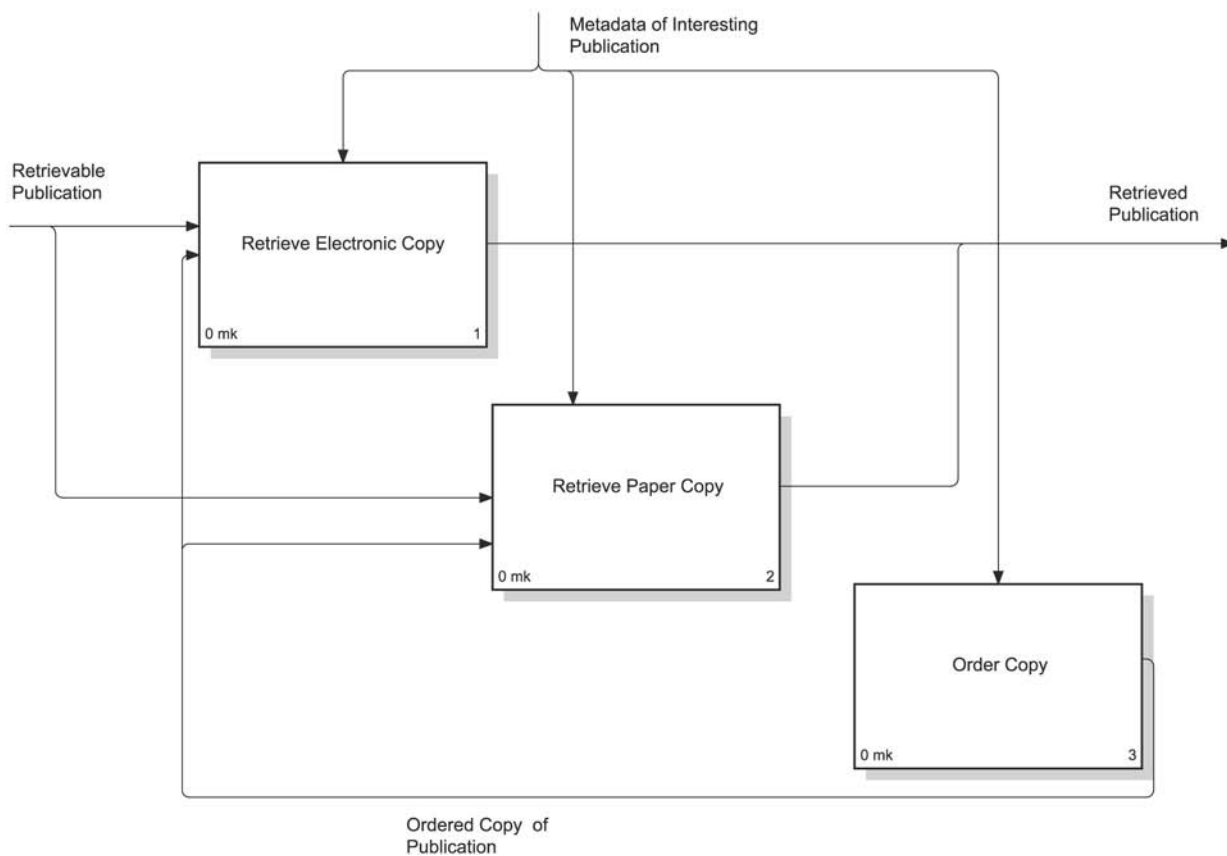


Figure 13 Diagram A32 – retrieve publication



organisation that the researcher belongs to does not subscribe to the journal in question, it might take quite a long time to obtain a copy of the article through some service for interlibrary loans or through a service for buying individual articles over the Web. Many younger researchers have become lazy about retrieving paper copies at all, since so much is on the Web.

Conclusions

The use of a formal process modelling language for a purpose such as this was motivated by the personal experience of the first author, having earlier used this type of methodology for the modelling of the construction process as well as having led an international project investigating innovative process modelling methodologies. The initial experiences have been very positive. The studied process is by its very nature rather linear (contrary to, for instance, architectural design), which makes the modelling easier than for processes involving a lot of networking or

iterative procedures. Also colleagues to whom the model has been shown have quite easily grasped the fundamentals of the IDEF0 notation and have been able to follow the logic of the model.

The model in its current shape is not yet validated in its details, but has been discussed with a number of domain experts (publishers, librarians) with encouraging feedback. Publishers employ methods of a similar nature to study the workflows within their organisations, but the point here is to study the whole process, including the activities of libraries and readers, and to use the process model as a basis for determining the activities which will be studied more in detail as a part of the cost modelling.

The cost modelling will be a synthesis task of data from several different sources. To some extent Web surveys will be used, in particular concerning the economics of open access journals and repositories and reader behaviour. Work of other researchers concerning the cost of certain activities in the model will also be

used (for example Tenopir and King, 1998; Odlyzko, 1998). The common denominator will be to study costs per publication flowing through the system. In addition to the cost modelling the model could also prove useful in providing a road map pointing out the position of a number of different initiatives for increasing access to scientific publications, within the overall system of scholarly communication.

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